Kyungrok Park

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Education

University of Toronto **Sep. 2019 – Apr. 2026**
Bachelor of Science in Statistical Science *Toronto ON Canada*

Relevant Coursework

- Time Series Analysis
- Statistical Methods for Machine Learning I & II
- Surveys, Samplings and Observational Data
- Stochastic Processes
- Method of Data Analysis
- Methods for Multivariate Data

Research Experience

UofT Staistical Sciences Directed Reading Program (DRP) **Jan 2026 - May 2026**

- Reviewed Research Papers on dimensionality reduction and randomized numerical linear algebra, including SVD, QR decomposition, subspace embedding, and LDA.
- Analyzed mathematical assumptions, algorithmic methods, and statistical applications presented in selected papers.
- Delivered technical presentations summarizing key ideas, results, and research contributions.

Projects

Randomized Linear Algebra and Subspace Embedding Simulation Study | *Python* **March 2026**

- Implemented simulation experiments comparing classical matrix factorization algorithms with randomized sketching approaches such as SRHT and Johnson–Lindenstrauss transforms
- Presented theoretical foundations of randomized projections, low-rank approximation, and subspace preservation through empirical simulation studies

Comparative Study of KV cache Quantization Methods | *Python* **May 2026 (in Progress)**

- Evaluated 2–8 bit KV-cache compression methods for transformer attention, including per-channel/token scaling, grouped residual caching, and symmetric/asymmetric quantization.
- Examined logit-level and multi-token tests to measure top-1 match, top-10 overlap, KL divergence, logit error, and compression ratio against full-precision KV-cache baselines.
- Identified that asymmetric per-channel key quantization with symmetric value quantization preserved prediction stability while achieving about 2× theoretical KV-cache compression.

An Empirical analysis of Energy intensity and Economic growth | *Stata* **December 2024**

- Develop a multiple regression model using economic data from 50 U.S. states, incorporating both state-fixed effects and year-fixed effects.
- Conduct a PCA analysis to determine the minimum number of factors required to effectively reduce the dimensionality of the data

Technical Skills

Languages: Python, SQL, Stata, R
Technologies/Frameworks: Latex, GitHub, Posit Cloud

Volunteer Work

Silim Nursery School **August 2018 – February 2019**
Assistant teacher for teaching Earth science *Seoul, South Korea*

- Make a simple experiment and booklet about Geology to teach students.
- Participate in exhibitions with students in Nursery School to motive them to improve their knowledge in geology

Christian Church **August 2018 – February 2019**
Assistant teacher for teaching Hanguel *Coquitlam, Vancouver*

- Assist in making booklet about Hanguel to teach 5-10 ages of children.
- Assist in teaching to help children build their language skills during a phase of their language development

Honors/Awards

UTM Entrance Awards | **November 2019**